

Response to Reviewers — BIOADV-2026-296 (revised resubmission)

We thank the reviewers and the Associate Editor for a rigorous and constructive review. The revision makes three structural changes before addressing the individual comments:

1. **Title and terminology.** The paper is retitled “*Near-duplicate leakage can reorder model rankings on a genomic benchmark suite: an audit of Genomic Benchmarks.*” We adopt **near-duplicate leakage** as the primary term (an exact- duplicate census shows the strongest dataset, `human_enhancers_ensembl`, is 38.0% *byte-identical* train/test copies), reserve *homology* for where alignment/containment evidence earns it, and rename the splitter **near-duplicate-aware** (addresses R3.4).
2. **Honest rescoping of the central claim.** The previous “changes which model wins on 2/2 leaky datasets” claim was too strong: on `human_nontata_promoters` the demotion holds only under accuracy@0.5 (it reverses under AUROC/F1, is a CI-overlap tie, and does not reproduce under a chromosome-holdout control). The ranking-change result is now claimed **cleanly for `human_enhancers_ensembl`** (an existence proof, airtight on every axis) and **as a cautionary partial case for `human_nontata_promoters`** (§4.3, Discussion).
3. **New experiments.** We added a from-scratch deep-model dose-response, an alignment-based re-measure, a chromosome-holdout control, a random-forest regularization path, a cluster/block bootstrap, a containment/canonical-k-mer/GC robustness battery, an imbalanced-prevalence panel, and an injected-leakage multiclass experiment. All numbers below are traceable to the results log.

Associate Editor (Prof. Shanfeng Zhu)

The Associate Editor summarized five overarching concerns; each is now addressed and cross-referenced to the detailed responses below.

- **Limited model scope.** A from-scratch 1D residual CNN with a pre-registered dropout×weight-decay dose-response now shows the effect reaches a trained neural network (regularized reference cell drops on both leaky datasets, CIs excluding zero); the named pretrained foundation models are scoped out with a *measured* ≈100% pretraining-overlap confound rather than merely asserted (R1.1, R2.a3, R3.1).
- **Validation of homology detection.** MMseqs2 alignment identity reproduces the effect metric-independently, and a length-robust containment index is added — which honestly flags one previously-clean dataset as borderline (R1.3, R2.a2, R3.4).
- **Bootstrap methodology.** A cluster (block) bootstrap over within-test near-duplicate components, with intraclass correlation, design effect, and an analytic cluster-robust cross-check, now quantifies the correlated-sample uncertainty; it changes no significance verdict (R3.3).

- **Dataset characteristics.** Class balance is disclosed as curated (not natural), and a prevalence-aware imbalanced evaluation is added (R2.a1).
- **Hyperparameter choice.** A random-forest regularization path shows the effect is a property of the *default unregularized* forest, dissolving under regularization (R3.2).

Most importantly, the central claim is now honestly rescoped from “changes which model wins on 2/2 leaky datasets” to a clean single-dataset existence proof plus an explicit cautionary partial case — directly addressing the generality-and-robustness concern at the heart of the decision.

Reviewer 1

R1.1 — “The evaluation uses only classical models; a trained deep network is needed, and the reach beyond this domain is unclear.”

(a) Deep model. We added a **from-scratch 1D residual CNN** (one-hot input, no k-mer features, trained on each training split only — no pretraining confound), with a **pre-registered** dropout×weight-decay dose-response and a binding refutation condition pre-specified in a timestamped document in the code release (results/deep_preregistration.md). The standard-practice regularized reference cell (dropout 0.3, weight decay 1e-4) drops on **both** leaky datasets — human_nontata_promoters **+0.043** (cluster-bootstrap CI [0.025, 0.061]), human_enhancers_ensembl **+0.014** ([0.007, 0.021]), both excluding zero — so the pre-registered refutation condition is **not** triggered. 19 of the 20 cells drop with cluster CIs excluding zero (the one exception is a single nontata cell, dropout 0.6/wd 1e-4, where heavy dropout underfits so there is no inflation to lose); per-cell single-run variance is non-trivial, so we read the mechanism from the grid-level contrast, and the unregularized manipulation-check cell memorizes hardest (ensembl graded gap **+0.222**, mirroring the unregularized RF). The effect is therefore **not** a classical-model artifact. New §4.6; Methods §3.8; pre-registration in results/deep_preregistration.md.

(b) Reach. We now separate two claims explicitly (Introduction): **Claim I** (de-leaking lowers accuracy) is credited to prior work across domains (Kapoor & Narayanan; hashFrag/Rafi; Bushuiev; PINDER) and reproduced only as a control; **Claim II** (leakage can *reorder* rankings) is our contribution, scoped to short human regulatory DNA plus a memorization-prone model. The title now carries that scope.

R1.2 — “The corrected ranking is treated as the true generalization ranking, but de-duplicating the training set also removes learnable signal.”

Agreed, and we now **measure** rather than assert this. On human_nontata_promoters the corrected and novel-only rankings diverge ($\tau = -0.67$): the random forest is demoted under the corrected split yet remains the best model on truly novel sequences (accuracy 0.879, <0.5 similarity), so its advantage is partly genuine generalization. We reframe both the corrected split and novel-only accuracy as explicit near-duplicate-bounded proxies (Methods §3.6/§3.7), name leave-one-chromosome-out as the functional ground truth (§4.8), and single-linkage cohesion (0.083 for nontata vs 0.96 for ensembl, §4.10) supplies the biological reason correction over-

removes signal on nontata. This is the core of the nontata-as- cautionary-case framing (§4.3, Discussion).

R1.3 — “Validate leakage with an alignment-based measure and give threshold recommendations.”

Alignment: we re-clustered both leaky datasets with **MMseqs2** alignment identity, fed the external clusters into the *same* whole-cluster splitter (only the edge metric changes), and refit all four models. The `human_enhancers_ensembl` effect is metric-independent (RF drop **0.164** Jaccard $\approx$ **0.162** alignment; RF still rank 4); nontata is milder under alignment ($0.108 \rightarrow 0.064$), reinforcing its partial status. We describe this as “alignment-scored,” not k-mer-independent, since MMseqs2 is k-mer-seeded (new §4.7, Methods §3.5). **Thresholds:** the clustering threshold sweep (0.5/0.7/0.9) changes no verdict (Methods §3.4); the length-robust containment re-measure and the bimodal-vs-graded similarity structure (cohesion 0.96 vs 0.083) provide the decision basis — a single blind-antimode threshold suffices for the discrete-duplication case (ensembl) whereas the chaining case (nontata) must be treated as partial-scope (§4.1, §4.10).

Reviewer 2

R2.a1 — “Class balance is not characterized, and imbalanced data are not tested.”

We now disclose that every balanced dataset is **curated** to a 0.500 positive fraction in both train and test (positive/negative subfolders); only nontata is a natural mild imbalance (0.544) (Methods §3.1). Because balance is preprocessed, we add a **prevalence stress test** (RF, $\pi \in \{0.5, 0.2, 0.1\}$) with prevalence-aware metrics: the leakage inflation is **revealed by AUPRC/MCC/minority-recall but masked by accuracy** (ensembl $\pi=0.2$: AUPRC $0.622 \rightarrow 0.477$, MCC $0.422 \rightarrow 0.182$, minority recall $0.258 \rightarrow 0.070$, accuracy barely moves $0.844 \rightarrow 0.808$). The prevalence-aware splitter preserves the target prevalence (realized 0.5001/0.2007/0.1001). New §4.11; Methods §3.10.

R2.a2 — “The similarity measure and the features are both k-mer based, so the leakage finding may be circular.”

Addressed by the alignment re-measure above (R1.3): with alignment identity as the edge metric and the same splitter, the `human_enhancers_ensembl` RF drop is essentially unchanged (0.164 vs 0.162) and the demotion persists, so the effect does not depend on the k-mer definition of similarity. We additionally note that ensembl’s leakage is 38.0% byte-identical duplication (metric-independent by construction). New §4.7; Methods §3.3/§3.5.

R2.a3 — “Extend to deep models and to the multiclass setting.”

Deep models: covered by the CNN (R1.1a, §4.6). **Multiclass:** the only multiclass set is clean, so we answer by construction — injecting label-carrying near-duplicates at fraction $f \in \{0, 0.1, 0.2, 0.4\}$ into the clean 3-class `human_ensembl_regulatory` set. The RF drop grows monotonically with f ($+0.0004, +0.118, +0.179, +0.258$) while LR stays flat (≤ 0.066), and the split removes it (corrected accuracy stable ≈ 0.58). New §4.12; Methods §3.11; we state explicitly that leakage is injected because the suite ships no naturally leaky multiclass set.

R2.b1 — “Give a threshold-sensitivity analysis and concrete recommendations.”

The verdict is robust to the clustering threshold (0.5/0.7/0.9 sweep, no change; Methods §3.4), and we now report the geometry that drives the recommendation: the length-robust containment index (§4.1/§4.10) as the metric to prefer when lengths vary, and single-linkage cohesion as the diagnostic that separates a clean discrete- duplication case (ensembl, treat with one threshold) from a transitive-chaining case (nontata, treat as partial-scope and avoid over-removal). Report-card guidance and the length-cap disclosure are in Table 2.

Reviewer 3

R3.1 — “Deep models are needed, specifically attention-based / foundation models (e.g. DNABERT-2, HyenaDNA).”

(i) Trained deep net: answered by a from-scratch 1D residual CNN (§4.6, one-hot input, no k-mer features, pre-registered dropout×weight-decay dose-response) trained on the training split only, whose regularized reference cell drops on both leaky datasets (cluster-bootstrap CIs excluding zero) — the memorization mechanism reaches a trained neural network with no pretraining confound. We also probed a from-scratch attention (Transformer) encoder under the identical protocol; consistent with transformers being data-hungry and weaker than convolutions on short, local-motif inputs, it underfits these 251–600 bp tasks and does not yield a clean aggregate-accuracy comparison, so we do not report it as evidence. The CNN is the appropriate trained-deep-net instrument here, and the pretraining-contamination argument below specifically justifies excluding *pretrained* attention models, not attention architectures per se. **(ii)/(iii) Named foundation models:** DNABERT-2, HyenaDNA, and the Nucleotide Transformer are cited and discussed but scoped **out** of the split-effect evidence, and we now **quantify** the confound rather than merely assert it: the two leaky datasets’ sequences are extracted from the human reference genome (we recovered their hg38 chromosomal coordinates for the chromosome-holdout control), so **≈100% of these test sequences already lie inside HyenaDNA’s whole-genome (hg38) pretraining corpus** and within the human component of DNABERT-2’s and the Nucleotide Transformer’s multispecies corpora — a second leakage channel that no train/test re-split can close. Fine-tuning a model that has already seen the test sequences during pretraining yields a confounded lower bound, not split-effect evidence, so we leave the deployed-foundation-model question **genuinely open** (Discussion, Limitations) rather than answer it with a contaminated number.

R3.2 — “The random forest runs at memorization-maximizing defaults; is the result just an untuned-RF artifact?”

We added a full **regularization path** ($\text{min_samples_leaf} \in \{1 \dots 500\}$, $\text{max_depth} \in \{\text{None} \dots 4\}$, at full scale, with the largest settings exceeding the largest near- duplicate cluster). At the default the forest scores 1.000 on the ≥ 0.9 -similarity bin with the full drop (0.164/0.108) and graded gap (0.230/0.121); regularizing collapses the drop to ≈ 0 ($0.164 \rightarrow -0.001$; $0.108 \rightarrow 0.000$) and the graded gap, and the “RF wins on the leaky split” phenomenon **requires** the unregularized forest (rf_rank_orig flips $1 \rightarrow 4$ once $\text{min_samples_leaf} \geq 20$). We frame this as an owned finding: the inflation and demotion are a property of the **default** high-capacity RF practitioners deploy, and

either regularization or a near-duplicate-aware split reveals the truth. New §4.5; Methods §3.7. The graded memorization gap is used as the confound-immune headline (immune to “regularized RF just got worse everywhere”).

R3.3 — “Near-duplicates violate independence; a cluster/block bootstrap is the correct uncertainty estimate.”

Implemented and run: resampling whole within-test near-duplicate components widens CIs by 1.0–1.5× but **flips no verdict** — the RF drop still excludes zero on both datasets (ensembl [0.156, 0.172]; nontata [0.093, 0.126]) and the ensembl-LR drop stays null. The design effect is small because the correlated near-duplicates are test-to-**train**, not test-to-test (within-test clustering only 1.06–1.19 sequences/component). New §4.9; Methods §3.9. Every deep-model drop CI (§4.6) already uses this cluster bootstrap. (The full design-effect/ICC/CRVE/combined-source detail is marked as being finalized in §3.9/§4.9.)

R3.4 — “‘Homology’ is the wrong word for near-identical/duplicate leakage.”

Agreed and adopted throughout: **near-duplicate leakage** is the primary term, the splitter is **near-duplicate-aware**, and the title is changed. An exact-duplicate census settles this metric-independently (human_enhancers_ensembl = 38.0% byte- identical copies; human_nontata_promoters = 0 exact but 22.5% near-duplicates at Jaccard ≥0.9). We keep “homology” only where the alignment/containment re-measure earns it (Methods §3.3), and we downgrade the report card’s green verdicts to “no forward-strand near-duplicate leakage detected (length-cap disclosed)” (Table 2, Discussion), since a clean verdict carries a heavier evidential burden than a leaky one.

Associate Editor — “This cannot be addressed in a short time / new experiments are needed.”

The revision adds the two hard-gate experiments the panel expected — a trained deep model (from-scratch CNN dose-response, §4.6) and alignment-based validation (MMseqs2, §4.7) — plus the most decisive genomics-standard control, a **leave-one-chromosome-out split** (§4.8), which independently reproduces the marquee human_enhancers_ensembl demotion (RF rank 1 → 4) and independently confirms that human_nontata_promoters is the partial case (RF stays rank 1). We also closed two internal issues a careful reviewer would have flagged: the central claim is now honestly rescoped to human_enhancers_ensembl (n=1 clean, partial on the second), and the previously inconsistent headline drop numbers are reconciled to a single decision rule (.predict()/argmax) and estimator (5-seed-mean + bootstrap CI), removing the earlier three-coexisting-values spread and the incorrect “thread-nondeterminism” wording.